Project Support Session - Personal Loan Campaign

Agenda

- Project Problem Statement and Dataset Overview
- Open Q&A
- Project Evaluation Rubric
- Project Submission Guidelines
- FAQs

Personal Loan Campaign - Business Context

AllLife Bank is a US bank that has a growing customer base. The majority of these customers are liability customers (depositors) with varying sizes of deposits. The number of customers who are also borrowers (asset customers) is quite small, and the bank is interested in expanding this base rapidly to bring in more loan business and in the process, earn more through the interest on loans. In particular, the management wants to explore ways of converting its liability customers to personal loan customers (while retaining them as depositors).

A campaign that the bank ran last year for liability customers showed a healthy conversion rate of over 9% success. This has encouraged the retail marketing department to devise campaigns with better target marketing to increase the success ratio.

Personal Loan Campaign - Objective

You as a Data Scientist at AllLife Bank have to build a model that will help the marketing department to identify the potential customers who have a higher probability of purchasing the loan.

To predict whether a liability customer will buy personal loans, to understand which customer attributes are most significant in driving purchases, and to identify which segment of customers to target more.

Personal Loan Campaign - Data Dictionary

Column Name	Description
ID	Customer ID
Age	Customer's age in completed years
Experience	# years of professional experience
Income	Annual income of the customer (in thousand dollars)
ZIP Code	Home Address ZIP code.
Family	The family size of the customer
CCAvg	Average spending on credit cards per month (in thousand dollars)
Education	Education Level. 1: Undergrad; 2: Graduate;3: Advanced/Professional

Personal Loan Campaign - Data Dictionary

Column Name	Description
Mortgage	Value of house mortgage if any. (in thousand dollars)
Personal_Loan	Did this customer accept the personal loan offered in the last campaign?
Securities_Account	Does the customer have a securities account with the bank?
CD_Account	Does the customer have a securities account with the bank?
Online	Does the customer have a certificate of deposit (CD) account with the bank?
CreditCard	Does the customer use a credit card issued by any other Bank (excluding All Life Bank)?

Personal Loan Campaign - Q&A

Personal Loan Campaign - Evaluation Rubric

Section	Points
Define the problem and perform an Exploratory Data Analysis	10
Data pre-processing	6
Model building - Decision Tree	10
Model Performance Evaluation and Improvement	20
Actionable Insights & Recommendations	6
Notebook Overall Quality	8

Personal Loan Campaign - Evaluation Rubric

Section 1: Define the problem and perform an Exploratory Data Analysis

What to Cover?	Guiding Questions and Considerations
Problem Definition & Questions	Clearly state the problem and define focused, answerable questions aligned with the dataset.
Data Background & Contents	Summarize dataset purpose, key features, and structure, noting missing values and data types.
Univariate Analysis	Explore each variable with suitable plots and highlight key patterns or distributions

Personal Loan Campaign - Evaluation Rubric

Section 1: Define the problem and perform an Exploratory Data Analysis

What to Cover?	Guiding Questions and Considerations
Bivariate Analysis	Examine relationships between variables using appropriate visualizations and summarize insights.
Provide comments/observations for the answers received	How do the responses look like? Are they too generic?

Personal Loan Campaign - Evaluation Rubric

Section 2: Data pre-processing

What to Cover?	Guiding Questions and Considerations
Missing Value Treatment	What would be the best methodology to handle missing values here? (impute by mean/median, impute by grouped mean/median, or something else?)
Outlier Detection & Treatment	What technique to use to detect unusual values? How do we handle such values? (remove, cap, or transform)
Feature Engineering	Are the existing features enough, or can we create a new feature that's likely to be impactful?

Personal Loan Campaign - Evaluation Rubric

Section 2: Data pre-processing

What to Cover?	Guiding Questions and Considerations
Data Preparation for Modelling	Do we need to do one-hot encoding? What proportion of the dataset should be considered for testing?

Personal Loan Campaign - Evaluation Rubric

Section 3: Model building - Decision Tree

What to Cover?	Guiding Questions and Considerations
Model Evaluation Criterion	Clearly define how you will measure model success (accuracy, precision, recall, etc.) based on the project's objectives.
Model Building & Performance	Why is the `random_state` parameter required here?
Decision Rules & Feature Importance	Visualize the tree structure and highlight the most important features driving predictions.

Personal Loan Campaign - Evaluation Rubric

Section 4: Model Performance Evaluation & Improvement

What to Cover?	Guiding Questions and Considerations
Model Improvement via Pruning	What values of `max_depth`, `min_samples_split`, and other hyperparameters would be effective? How to find the optimal value of `ccp_alpha`?
Performance Comparison & Final Model Selection	Compare all models' performance metrics and select the best-performing model for the project.
Decision Rules & Feature Importance	Extract and interpret decision rules from the final model and highlight key features influencing predictions.

Personal Loan Campaign - Evaluation Rubric

Section 5: Actionable Insights and Recommendations

What to Cover?	Guiding Questions and Considerations
Conclude with the key takeaways for the business	What are the key things that we learned from the data that's useful for the business? Logical next steps are a key inclusion

Personal Loan Campaign - Evaluation Rubric

Section 6: Notebook - Overall Quality

What to Cover?	Guiding Questions and Considerations
Structure and flow	Clear structure, flow, and visual appeal - everything sits well in a story.
Well-commented code	Comments clear, and helpful in explaining the logic, intent, and flow of the code.
All code executed and necessary output visible	Code runs without errors, and are all required outputs are clearly displayed in the right format and context.
No Errors	Code execute smoothly without runtime, syntax, or logical errors that could break functionality.

Personal Loan Campaign - Project Submission Guidlines

Download the dataset, write necessary code in a Python notebook to solve the questions, and perform all the tasks as per the grading rubric and insight-based questions.

Clearly write down observations, insights, and recommendations for the business problem based on the analysis performed

Once the notebook is complete and all the sections mentioned in the grading rubric have been added, download it as a **.ipynb** file and convert it to a **.html** file

The conversion can be done via one of the following ways:

Jupyter Notebook: ***Download as HTML*** from the ***File*** menu
Google Colab: Use [free online tools]

Personal Loan Campaign - FAQ

Decision Tree arrows are missing. How to fix this?

Use the following code as a reference to resolve the issue and make necessary changes (name of the model, feature names, etc):

```
plt.figure(figsize=(20,30))
out = tree.plot_tree(model,feature_names=feature_names,filled=True,fontsize=9,node_ids=False,class_names=None,)
#below code will add arrows to the decision tree split if they are missing
for o in out:
    arrow = o.arrow_patch
    if arrow is not None:
        arrow.set_edgecolor('black')
        arrow.set_linewidth(1)
plt.show()
```

Personal Loan Campaign - FAQ

How to deal with "ZIPCode" as it is a numeric value but it's also essentially a category?

You can explore the following links to deal with zip codes:

uszipcode- Python package that can help in mapping zip codes to different locations

https://www.smarty.com/articles/zip-4-code - Description of how zip codes are created in the US.

Personal Loan Campaign - FAQ

Should I create dummies for the columns that only have 0’s and 1’s?

No, it is not necessary to create dummies for these columns.

Personal Loan Campaign - FAQ

I'm trying to post-prune the decision tree. But I'm getting the following error:

```
"ValueError: ccp_alpha must be greater than or equal to 0"
```

How to resolve this?

To resolve this error kindly use absolute values (positive value) of alpha. Use the following lines of code to resolve the error:

```
ccp_alphas, impurities = abs(path.ccp_alphas), path.impurities
```

Happy Learning !